9.5.2 Summary ¶

Definition 9.5.2.1. Eigenvalue, eigenvector, and eigenpair. Let $A \in \mathbb{C}^{m\times m}$. Then $\lambda \in \mathbb{C}$ and nonzero $x \in \mathbb{C}^m$ are said to be an eigenvalue and corresponding eigenvector if $Ax = \lambda x$. The tuple (λ, x) is said to be an eigenpair.

For $A \in \mathbb{C}^{m\times m}$, the following are equivalent statements:

- A is nonsingular.
- A has linearly independent columns.
- There does not exist $x \neq 0$ such that $Ax = 0$.
- $\mathcal{N}(A) = \{0\}$. (The null space of A is trivial.)
- $\dim(\mathcal{N}(A)) = 0$.
- $\det(A) \neq 0$.

For $A \in \mathbb{C}^{m\times m}$, the following are equivalent statements:

- λ is an eigenvalue of A
- $(\lambda I - A)$ is singular.
- $(\lambda I - A)$ has linearly dependent columns.
- There exists $x \neq 0$ such that $(\lambda I - A)x = 0$.
- The null space of $\lambda I - A$ is nontrivial.
- $\dim(\mathcal{N}(\lambda I - A)) > 0$.
- $\det(\lambda I - A) = 0$.

Definition 9.5.2.2. Spectrum of a matrix. The set of all eigenvalues of A is denoted by $\Lambda(A)$ and is called the spectrum of A.

Definition 9.5.2.3. Spectral radius. The spectral radius of A, $\rho(A)$, equals the magnitude of the largest eigenvalue, in magnitude:

$$\rho(A) = \max_{\lambda \in \Lambda(A)} |\lambda|.$$

Theorem 9.5.2.4. Gershgorin Disk Theorem. *Let $A \in \mathbb{C}^{m\times m}$,*

$$A = \begin{pmatrix} \alpha_{0,0} & \alpha_{0,1} & \cdots & \alpha_{m-1,m-1} \\ \alpha_{0,0} & \alpha_{0,1} & \cdots & \alpha_{m-1,m-1} \\ \vdots & \vdots & & \vdots \\ \alpha_{0,0} & \alpha_{0,1} & \cdots & \alpha_{m-1,m-1} \end{pmatrix},$$

$$\rho_i(A) = \sum_{j\neq i} |\alpha_{i,j}|,$$

and

$$R_i(A) = \{x \text{ s.t. } |x - \alpha_{i,i}| \leq \rho_i\}.$$

In other words, $\rho_i(A)$ equals the sum of the absolute values of the off diagonal elements of A in row i and $R_i(A)$ equals the set of all points in the complex plane that are within a distance ρ_i of diagonal element $\alpha_{i,i}$. Then

$$\Lambda(A) \subset \cup_i R_i(A).$$

In other words, every eigenvalue lies in one of the disks of radius $\rho_i(A)$ around diagonal element $\alpha_{i,i}$.

Corollary 9.5.2.5. *Let A and $R_i(A)$ be as defined in Theorem 9.5.2.4. Let K and K^C be disjoint subsets of $\{0,\ldots,m-1\}$ such that $K \cup K^C = \{0,\ldots,m-1\}$. In other words, let K be a subset of $\{0,\ldots,m-1$ and K^C its complement. If*

$$\left(\cup_{k\in K} R_k(A)\right) \cap \left(\cup_{j\in K^C} R_j(A)\right) = \emptyset$$

then $\cup_{k\in K} R_k(A)$ contains exactly $|K|$ eigenvalues of A (multiplicity counted). In other words, if $\cup_{k\in K} R_k(A)$ does not intersect with any of the other disks, then it contains as many eigenvalues of A (multiplicity counted) as there are elements of K.

Some useful facts for $A \in \mathbb{C}^{m\times m}$:

- $0 \in \Lambda(A)$ if and only A is singular.
- The eigenvectors corresponding to distinct eigenvalues are linearly independent.
- Let A be nonsingular. Then (λ, x) is an eigenpair of A if and only if $(1/\lambda, x)$ is an eigenpair of A^{-1}.
- (λ, x) is an eigenpair of A if and only if $(\lambda - \rho, x)$ is an eigenpair of $A - \rho I$.

Some useful facts for Hermitian $A \in \mathbb{C}^{m\times m}$:

- All eigenvalues are real-valued.
- A is HPD if and only if all its eigenvalues are positive.
- If (λ, x) and (μ, y) are both eigenpairs of Hermitian A, then x and y are orthogonal.

Definition 9.5.2.6. The determinant of

$$A = \begin{pmatrix} \alpha_{00} & \alpha_{01} \\ \alpha_{10} & \alpha_{11} \end{pmatrix}$$

is given by

$$\det(A) = \alpha_{00}\alpha_{11} - \alpha_{10}\alpha_{01}.$$

The characteristic polynomial of

$$A = \begin{pmatrix} \alpha_{00} & \alpha_{01} \\ \alpha_{10} & \alpha_{11} \end{pmatrix}$$

is given by

$$\det(\lambda - IA) = (\lambda - \alpha_{00})(\lambda - \alpha_{11}) - \alpha_{10}\alpha_{01}.$$

This is a second degree polynomial in λ and has two roots (multiplicity counted). The eigenvalues of A equal the roots of this characteristic polynomial.

The characteristic polynomial of $A \in \mathbb{C}^{m\times m}$ is given by

$$\det(\lambda - IA).$$

This is a polynomial in λ of degree m and has m roots (multiplicity counted). The eigenvalues of A equal the roots of this characteristic polynomial. Hence, A has m eigenvalues (multiplicity counted).

Definition 9.5.2.7. Algebraic multiplicity of an eigenvalue. Let $A \in \mathbb{C}^{m\times m}$ and $p_m(\lambda)$ its characteristic polynomial. Then the (algebraic) multiplicity of eigenvalue λ_i equals the multiplicity of the corresponding root of the polynomial.

If

$$p_m(\chi) = \alpha_0 + \alpha_1\chi + \cdots + \alpha_{m-1}\chi^{m-1} + \chi^m$$

and

$$A = \begin{pmatrix} -\alpha_{n-1} & -\alpha_{n-2} & -\alpha_{n-3} & \cdots & -\alpha_1 & -\alpha_0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

then

$$p_m(\lambda) = \det(\lambda I - A).$$

Corollary 9.5.2.8. *If $A \in \mathbb{R}^{m\times m}$ is real-valued then some or all of its eigenvalues may be complex-valued. If eigenvalue λ is complex-valued, then its conjugate, $\bar{\lambda}$, is also an eigenvalue. Indeed, the complex eigenvalues of a real-valued matrix come in complex pairs.*

Corollary 9.5.2.9. *If $A \in \mathbb{R}^{m\times m}$ is real-valued and m is odd, then at least one of the eigenvalues of A is real-valued.*

Let λ be an eigenvalue of $A \in \mathbb{C}^{m\times m}$ and

$$\mathcal{E}_\lambda(A) = \{x \in \mathbb{C}^m | Ax = \lambda x\}.$$

bet the set of all eigenvectors of A associated with λ plus the zero vector (which is not considered an eigenvector). This set is a subspace.

The elements on the diagonal of a diagonal matrix are its eigenvalues. The elements on the diagonal of a triangular matrix are its eigenvalues.

Definition 9.5.2.10. Given a nonsingular matrix Y, the transformation $Y^{-1}AY$ is called a similarity transformation (applied to matrix A).

Let $A, B, Y \in \mathbb{C}^{m\times m}$, where Y is nonsingular, $B = Y^{-1}AY$, and (λ, x) an eigenpair of A. Then $(\lambda, Y^{-1}x)$ is an eigenpair of B.

Theorem 9.5.2.11. *Let $A, Y, B \in \mathbb{C}^{m\times m}$, assume Y is nonsingular, and let $B = Y^{-1}AY$. Then $\Lambda(A) = \Lambda(B)$.*

Definition 9.5.2.12. Given a nonsingular matrix Q the transformation Q^HAQ is called a unitary similarity transformation (applied to matrix A).

Theorem 9.5.2.13. Schur Decomposition Theorem. *Let $A \in \mathbb{C}^{m\times m}$. Then there exist a unitary matrix Q and upper triangular matrix U such that $A = QUQ^H$. This decomposition is called the Schur decomposition of matrix A.*

Theorem 9.5.2.14. Spectral Decomposition Theorem. *Let $A \in \mathbb{C}^{m\times m}$ be Hermitian. Then there exist a unitary matrix Q and diagonal matrix $D \in \mathbb{R}^{m\times m}$ such that $A = QDQ^H$. This decomposition is called the spectral decomposition of matrix A.*

Theorem 9.5.2.15. *Let $A \in \mathbb{C}^{m\times m}$ be of form $A = \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline 0 & A_{BR} \end{array}\right)$, where A_{TL} and A_{BR} are square submatrices. Then $\Lambda(A) = \Lambda(A_{TL}) \cup \Lambda(A_{BR})$.*

Definition 9.5.2.16. Diagonalizable matrix. A matrix $A \in \mathbb{C}^{m \times m}$ is said to be diagonalizable if and only if there exists a nonsingular matrix X and diagonal matrix D such that $X^{-1}AX = D$.

Theorem 9.5.2.17. *A matrix $A \in \mathbb{C}^{m \times m}$ is diagonalizable if and only if it has m linearly independent eigenvectors.*

If $A \in \mathbb{C}^{m \times m}$ has distinct eigenvalues, then it is diagonalizable.

Definition 9.5.2.18. Defective matrix. A matrix $A \in \mathbb{C}^{m \times m}$ that does not have m linearly independent eigenvectors is said to be defective.

Corollary 9.5.2.19. *Matrix $A \in \mathbb{C}^{m \times m}$ is diagonalizable if and only if it is not defective.*

Definition 9.5.2.20. Geometric multiplicity. Let $\lambda \in \Lambda(A)$. Then the geometric multiplicity of λ is defined to be the dimension of $\mathcal{E}_\lambda(A)$ defined by

$$\mathcal{E}_\lambda(A) = \{x \in \mathbb{C}^m | Ax = \lambda x\}.$$

In other words, the geometric multiplicity of λ equals the number of linearly independent eigenvectors that are associated with λ.

Definition 9.5.2.21. Jordan Block. Define the $k \times k$ Jordan block with eigenvalue μ as

$$J_k(\mu) = \begin{pmatrix} \mu & 1 & 0 & \cdots & 0 & 0 \\ 0 & \mu & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ddots & \mu & 1 \\ 0 & 0 & 0 & \cdots & 0 & \mu \end{pmatrix}$$

Theorem 9.5.2.22. Jordan Canonical Form Theorem. *Let the eigenvalues of $A \in \mathbb{C}^{m \times m}$ be given by $\lambda_0, \lambda_1, \cdots, \lambda_{k-1}$, where an eigenvalue is listed as many times as its geometric multiplicity. There exists a nonsingular matrix X such that*

$$X^{-1}AX = \left(\begin{array}{c|c|c|c} J_{m_0}(\lambda_0) & 0 & \cdots & 0 \\ \hline 0 & J_{m_1}(\lambda_1) & \cdots & 0 \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & \cdots & J_{m_{k-1}}(\lambda_{k-1}) \end{array}\right).$$

A practical Power Method for finding the eigenvector associated with the largest eigenvalue (in magnitude):

Pick $v^{(0)}$ of unit length
for $k = 0, \ldots$
  $v^{(k+1)} = Av^{(k)}$
  $v^{(k+1)} = v^{(k+1)} / \|v^{(k+1)}\|$
endfor

Definition 9.5.2.23. Rayleigh quotient. If $A \in \mathbb{C}^{m \times m}$ and $x \neq 0 \in \mathbb{C}^m$ then

$$\frac{x^H Ax}{x^H x}$$

is known as the ***Rayleigh quotient***.

If x is an eigenvector of A, then

$$\frac{x^H Ax}{x^H x}$$

is the associated eigenvalue.

Definition 9.5.2.24. Convergence of a sequence of scalars. Let $\alpha_0, \alpha_1, \alpha_2, \ldots \in \mathbb{R}$ be an infinite sequence of scalars. Then α_k is said to converge to α if

$$\lim_{k \to \infty} |\alpha_k - \alpha| = 0.$$

Definition 9.5.2.25. Convergence of a sequence of vectors. Let $x_0, x_1, x_2, \ldots \in \mathbb{C}^m$ be an infinite sequence of vectors. Then x_k is said to converge to x if for any norm $\| \cdot \|$

$$\lim_{k \to \infty} \|x_k - x\| = 0.$$

Definition 9.5.2.26. Rate of convergence. Let $\alpha_0, \alpha_1, \alpha_2, \ldots \in \mathbb{R}$ be an infinite sequence of scalars that converges to α. Then

- α_k is said to converge linearly to α if for sufficiently large k

$$|\alpha_{k+1} - \alpha| \leq C|\alpha_k - \alpha|$$

  for some constant $C < 1$.
- α_k is said to converge superlinearly to α if for sufficiently large k

$$|\alpha_{k+1} - \alpha| \leq C_k|\alpha_k - \alpha|$$

with $C_k \to 0$.

- α_k is said to converge quadratically to α if for sufficiently large k

$$|\alpha_{k+1} - \alpha| \le C|\alpha_k - \alpha|^2$$

for some constant C.

- α_k is said to converge superquadratically to α if for sufficiently large k

$$|\alpha_{k+1} - \alpha| \le C_k|\alpha_k - \alpha|^2$$

with $C_k \to 0$.

- α_k is said to converge cubically to α if for large enough k

$$|\alpha_{k+1} - \alpha| \le C|\alpha_k - \alpha|^3$$

for some constant C.

The convergence of the Power Method is linear.

A practical Inverse Power Method for finding the eigenvector associated with the smallest eigenvalue (in magnitude):

$$\begin{array}{l} \text{Pick } v^{(0)} \text{ of unit length} \\ \mathbf{for}\ k = 0, \ldots \\ \quad v^{(k+1)} = A^{-1} v^{(k)} \\ \quad v^{(k+1)} = v^{(k+1)} / \| v^{(k+1)} \| \\ \mathbf{endfor} \end{array}$$

The convergence of the Inverse Power Method is linear.

A practical Rayleigh quation iteration for finding the eigenvector associated with the smallest eigenvalue (in magnitude):

$$\begin{array}{l} \text{Pick } v^{(0)} \text{ of unit length} \\ \mathbf{for}\ k = 0, \ldots \\ \quad \rho_k = v^{(k)H} A v^{(k)} \\ \quad v^{(k+1)} = (A - \rho_k I)^{-1} v^{(k)} \\ \quad v^{(k+1)} = v^{(k+1)} / \| v^{(k+1)} \| \\ \mathbf{endfor} \end{array}$$

The convergence of the Rayleigh Quotient Iteration is quadratic (eventually, the number of correct digits doubles in each iteration). If A is Hermitian, the convergence is cubic (eventually, the number of correct digits triples in each iteration).